Hydrostatics
(Liquids at Rest)

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Hydrostatics

Fluid:
A fluid is a substance which deforms continuously on application of a shear force. When a fluid is subjected to a shear force, the layers of fluid slide relative to each other.
Fluid has the ability to flow or change its shape under the application of force. Liquids and gases or a mixture of both are fluids.

Properties of fluid :

1. **Density:** Density is quantity of matter contained in a unit volume of substance.

   Density, (ρ) $\quad \rho = \lim_{\Delta v \to 0}\left(\frac{\Delta m}{\Delta v}\right)$

   Dimensions of density = $[ML^{-3}]$

   SI unit = kg/m^3

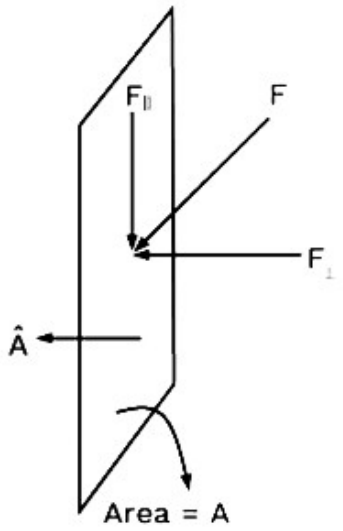
2. **Relative Density:** For solids and liquids, relative density is the ratio of density of material to the density of water at (4°C)

   $$R.D. = \frac{\text{Density of substane}}{\text{Density of water}} = \frac{\rho}{\rho_w}$$

   For gases, relative density is the ratio of density of the gas to the density of air.

3. **Pressure**: It is defined as the magnitude of normal force acting on a unit surface area.

   $$P = \lim_{A \to 0} \frac{F_\perp}{A}$$



In vector form,

$$\vec{F} = P\vec{A}$$

⇒ Pressure is a scalar quantity.

Pressure at a point acts in all directions and a particular direction cannot be associated with it.

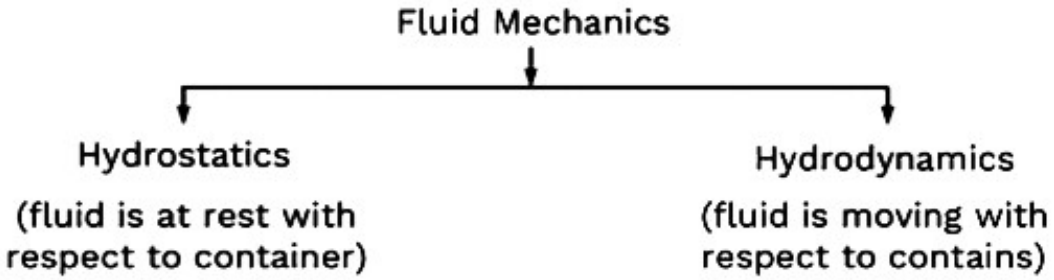
⇒ SI unit:– Pascal

$$1 \text{ Pascal} = \frac{1N}{m^2}$$

⇒ CGS unit:– $1 dyne/cm^2$

⇒ Dimension:- $[P] = \frac{[F]}{[A]} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$

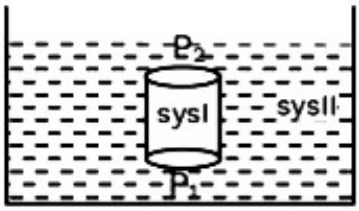
⇒ $[P] = M^1L^{-1}T^{-2}$



Hydrostatics is study of fluid when it is at rest with respect to the container

Variation of pressure inside a fluid with depth and height

Let's consider a small volume of element having height h and area of cross section A, upper surface pressure P_2 and lower surface pressure P_1



F.B.D. of system I

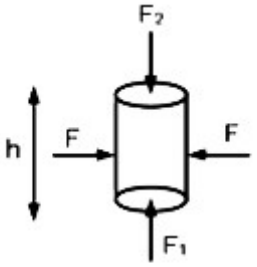
$[P_2 = F_2/A]$

$[P_1 = F_1/A]$

$F_1 - F_2 = mg$

$P_1A - P_2A = (\rho Ah)g \left[\because \ m = \rho V = \rho Ah\right]$

$$\boxed{P_1 - P_2 = \rho gh}$$



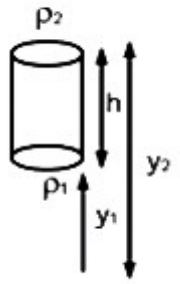
Pressure variation in differential form

$P_1 - P_2 = \rho g(g_2 - y_1)$

$P_1 - P_2 = -\rho g\,(y_2 - y_1)$

$\Delta P = -\rho g \Delta y$

$$\boxed{\frac{dP}{dy} = -\rho g}$$



⇒ Negative sign indicates pressure decreases with y coordinate

Q. **If pressure at the half depth of lake is equal to 2/3 times at the bottom of lake then find the depth of lake? (take density of water (ρ) = $10^3 kg/m^3$)**

Sol.

$P_A = (P_0 + \rho gd/2)$

$P_B = (P_0 + \rho gd)$

$$(P_A) = \frac{2}{3}(P_B)$$

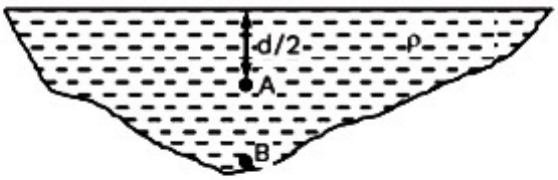
$$\Rightarrow \quad P_0 + \rho g\left(\frac{d}{2}\right) = \frac{2}{3}(P_0 + \rho gd)$$

$$\Rightarrow \quad P_0 + \frac{\rho gd}{2} = \frac{2}{3}P_0 + \frac{2}{3}\rho gd$$

$$\Rightarrow \quad P_0 = \frac{\rho gd}{2}$$

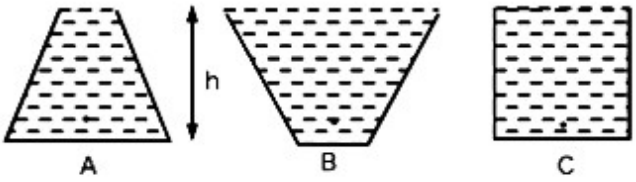
$$\Rightarrow \quad d = \frac{2P_0}{\rho g}$$

$$\Rightarrow \quad d = 20 \text{ m}$$



Note:

- Fluid pressure does not depend on shape of container but it only depends on depth from free surface
- The absolute pressure at depth h below the free liquid surface is more than the pressure at free surface by an amount ρgh.



$P_A = P_B = P_C = P_0 + \rho gh$ (where P_0 is pressure at free surface)

Q. **Find pressure at different points that is at A, B, C, D.**
Given: $P_A = 10KPa$
$\rho_w = 10^3 Kg/m^3$

Sol.

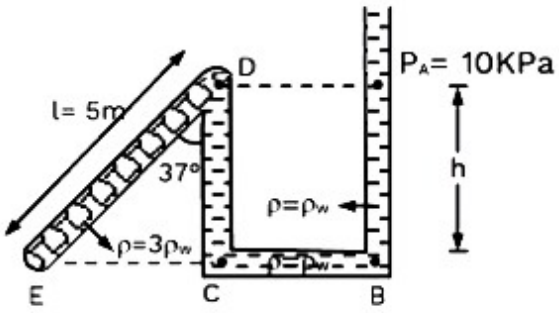
$P_B = Pa + \rho_w gh$
$= 10KPa + 10^3 \times 10 \times (l \cos 37°)$
$= 10\ KPa + 10^3 \times 10 \times 4$
$= 50\ KPa$

$P_C = 50\ KPa$

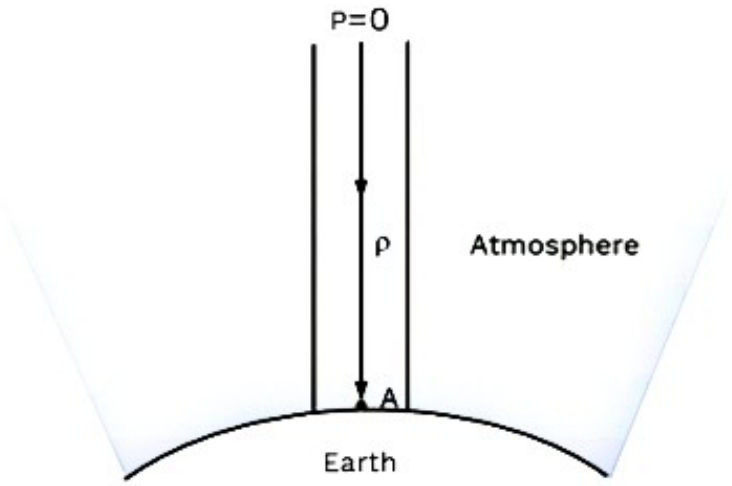
$P_D = 10\ KPa$

$P_E = P_B + (3\rho_w)(g)(l \cos 37°)$
$= 10\ KPa + 3 \times 10^3 \times 10 \times 4$

$P_E = 130\ KPa$



Atmospheric Pressure:
Atmospheric pressure is equal to the hydrostatic pressure caused by the weight of air above the measurement point.



$P_A = 0 + \int \rho g dy$

$P_A = \int \rho g dy$

$P_A = 1.013 \times 10^5$ Pa

$= 76$ cm of Hg

$= 760$ mm of Hg

$= 760$ torr

$= 1$ bar

$= 1$ atm pressure

Note: As elevation increases, there is less overlying atmospheric mass, that's why atmospheric pressure decreases with increasing elevation.

Mercury Barometer: It was developed by *Evangelista Torricelli.*

It measures the atmospheric pressure with the adjustment of mercury in the tube until atmospheric force exerted on the resevoir is balanced by the weight of the mercury column.

The barometer contains a vertical glass tube which is closed at one end.

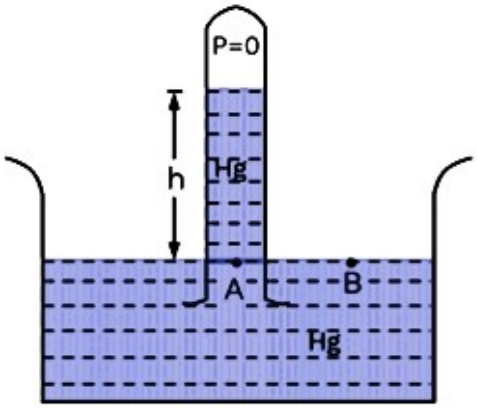
A vertical long and hollow glass is tube is dipped in mercury reservoir with air evacuated from it.

$P_A = P_B$ = Atmospheric pressure

$\rho_{Hg} \times g \times h = 101.32$ KPa

$$\Rightarrow h = \frac{101.32 \times 10^3}{13.5 \times 10^3 kg/m^2 \times 10}$$

$\Rightarrow h = 76$ cm



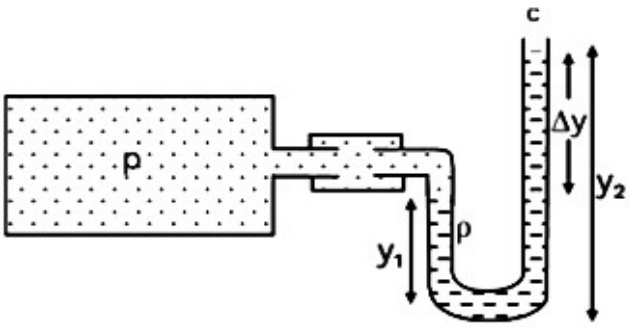
Height of water in barometer

$\rho_w gh = 101.325$KPa

$$\Rightarrow h = \frac{101.32 \times 10^3}{10^3 \times 10} = 10m$$

Open Tube Manometer:

It consists of a vertical glass tube, the lower end of which is attached to the point where the pressure needs to be measured and the upper end is left open to the atmosphere. It works on a very simple principle. The height of liquid in the tube balances the pressure in the container to which it is connected.



$P = P_0 + \rho g \Delta y$

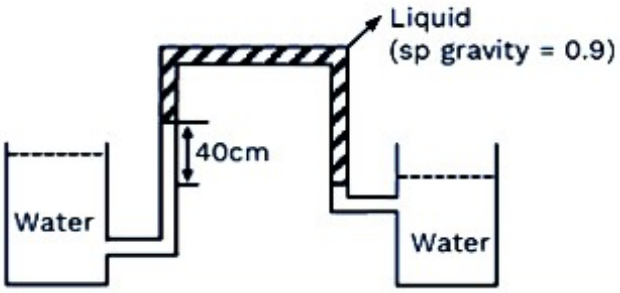
$P = P_0 + \rho g(y_2 - y_1)$

$P - P_0 = \rho g \Delta y$ = Gauge pressure

Gauge Pressure is the difference between absolute pressure (P_{abs}) and the prevailing atmospheric pressure (P_0) and is given by:

$$P_g = P_{abs} - P_0$$

Q. **The manometer shown in the following figure is used to measure the water level difference between the two tanks. Find the water level difference for the conditions shown.**



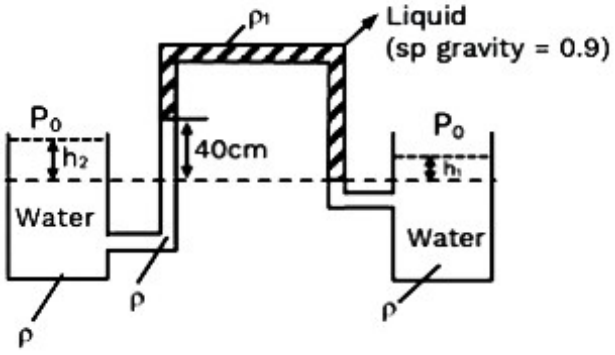
Sol. $P_0 + h_1\rho g - 40\rho_1 g + 40\rho g = P_0 + h_2\rho g$

$h_2\rho g - h_1\rho g = 40\rho g - 40\,\rho_1 g$

as $\rho_1 = 0.9\,\rho$

$(h_2 - h_1)\rho g = 40\rho g - 36\rho g$

$h_2 - h_1 = 4$ cm



PASCAL'S LAW:

As per Pascal's law, whenever an external pressure is applied to any part of a fluid kept at rest in a vessel, it is transmitted undiminished and equally in all directions.

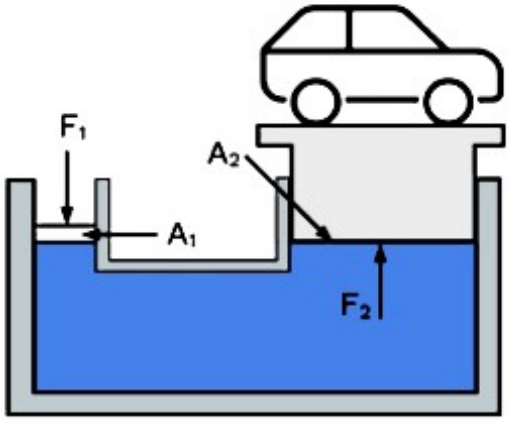
Hydraulic lift:

This can be used to lift a heavy load placed on the platform of larger piston or to press the thing placed between the piston and the heavy platform.

$P_1 = P_2$

$\Rightarrow \dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$

$\Rightarrow F_1 = \dfrac{F_2}{A_2} \times A_1 \quad \Rightarrow \quad F_1 = \dfrac{W}{A_2} \times A_1$

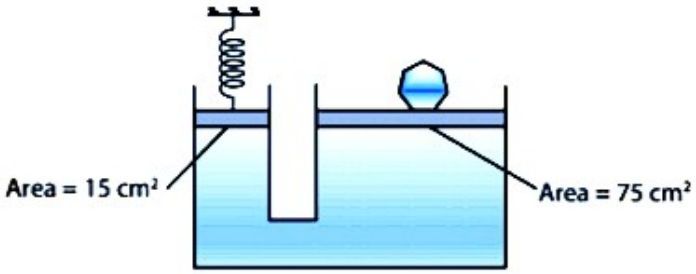


As $A_1 << A_2$, $F_1 << W$

So only a small amount of force(F_1) is required to lift heavy car(W) through hydraulic lift

Note: The work done by applied force is equal to change in potential energy of the weight in hydraulic lift.

Q. **The drawing shows a hydraulic chamber with a spring (spring constant = 1600 N/m) attached to the input piston and a rock of mass 40.0 kg resting on the output plunger. The piston and plunger are nearly at the same height, and each has a negligible mass. By how much is the spring compressed from its unstrained position?**



Sol. According to pascal's law:

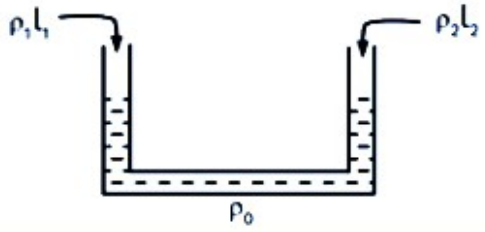
$$F_1 = F_2\left(\frac{A_1}{A_2}\right) = mg\left(\frac{A_1}{A_2}\right) \quad \text{...(i)}$$

$$\text{Also, } x = \left(\frac{F_1}{k}\right) \quad \text{...(ii)}$$

Substituting Eq. (i) into Eq. (ii), we find that

$$x = \frac{F_1}{k} = \frac{mg}{k}\left(\frac{A_1}{A_2}\right) = \frac{(40\text{kg})(10\ \text{m/s}^2)}{1600\ \text{N/m}}\left(\frac{15\ \text{cm}^2}{75\ \text{cm}^2}\right) = 5.0 \times 10^{-2}\text{m}$$

Q. **Two liquids having density ρ_1 and ρ_2 are added in 2 columns of tube as shown, length of column added are l_1 and l_2. Find x through which liquid of density ρ_0 will rise**



Sol. $P_A = P_0 + \rho_1 l_1 g$

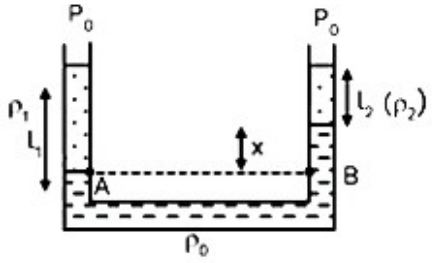
$P_B = P_0 + \rho_2 l_2 g + \rho_0 x g$

$P_a = P_B$

$$P_0 + \rho_1 l_1 g = P_0 + \rho_2 l_2 g + \rho_0 x g$$

$$\rho_1 l_1 = \rho_2 l_2 + \rho_0 x$$

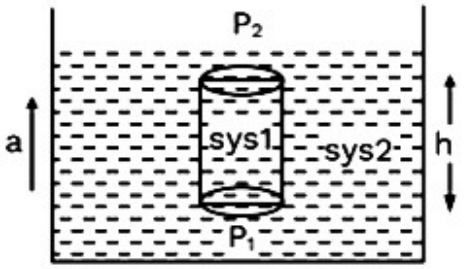
$$\Rightarrow \quad x = \frac{\rho_1 l_1 - \rho_2 l_2}{\rho_0}$$



Pressure difference in an uniformly Accelerated fluid

Case-1: Vertical acceleration:

Consider a cylindrical element of height h and area A in a container moving in vertical direction with acceleration a



From free body diagram of system 1

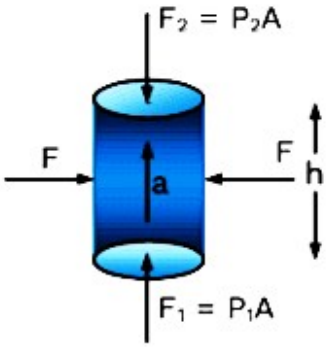
$F_1 - F_2 - mg = ma$

$F_1 - F_2 = mg + ma$

$[m = \rho v = \rho Ah]$

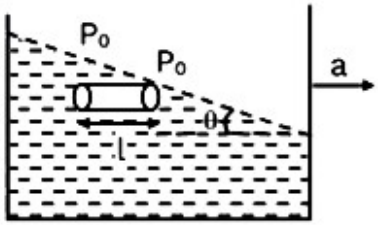
$(P_1 - P_2)A = (\rho Ah)(g + a)$

$P_1 - P_2 = \rho h(g + a)$



Case-2: Horizontal Acceleration:

Consider a cylindrical element length l and area A in a container moving in horizontal direction with acceleration a as shown below.



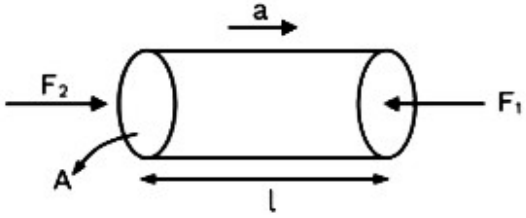
Now by making free body diagram of given element from case-2.

$F_2 - F_1 = ma$

$P_1A - P_1A = \rho(lA)a$

$P_2 - P_1 = \rho la$

$P_2 - P_1 = \rho la$



Calculation of Angle θ

$P_A = P_0 + \rho x a$

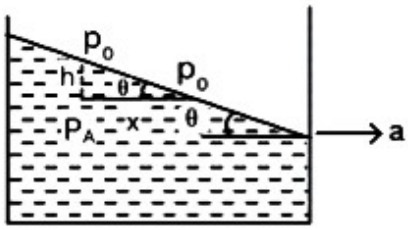
$P_A = P_0 + \rho h g$

From above equation

$\rho x a = \rho h g$

$\Rightarrow \frac{h}{x} = \frac{a}{g}$

$\Rightarrow \tan\theta = \frac{a}{g}$

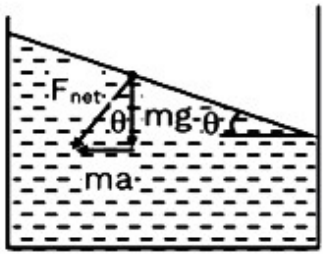


Alternatively

Net force on water molecules at surface would be perpendicular to water surface. From vector triangle.

$\tan\theta = \left(\frac{ma}{mg}\right)$

$\Rightarrow \tan\theta = a/g$



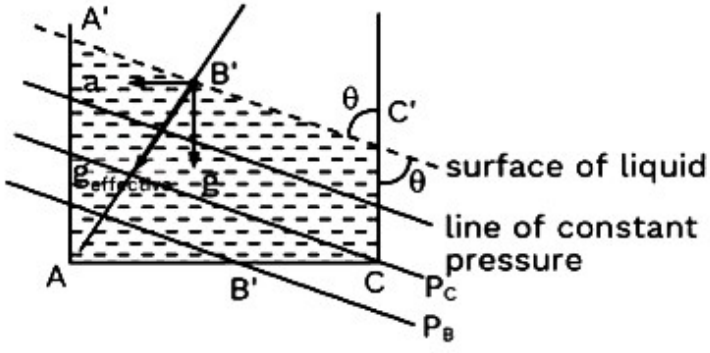
Note: Line of constant pressure would be parallel to surface of liquid

$F_{net} = \sqrt{(mg)^2 + (ma)^2}$

$= m\sqrt{g^2 + a^2}$

$= m\, g_{eff}$

$g_{eff} = \sqrt{a^2 + g^2}$



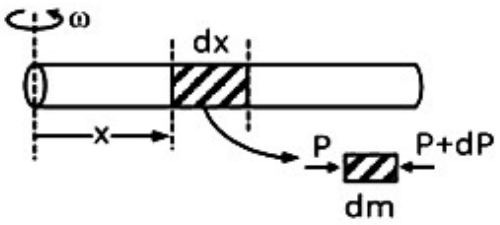
Pressure increases along the line of g_{eff}

$P = P_0 + \rho g_{eff} h$

& $P_A > P_B > P_C$

Pressure difference in rotating fluid:

Suppose that a liquid of density ρ kept inside a tube of area of cross-section A is rotating with an angular velocity 'ω' as shown.



Consider a small element of fluid having length 'dx' at a distance x from the axis of rotation. Mass of this element is,

$dm = (\rho A dx)$

This element is rotating in a circle of radius 'x'. So, this is accelerated towards centre with a centripetal acceleration.

$a = x\omega^2$ (as $a = r\omega^2$)

To provide this acceleration, pressure on right hand side of the element should be more.

$\therefore \quad (P + dP)A - (P)A = (dm)a = (\rho A dx)(d\omega^2)$

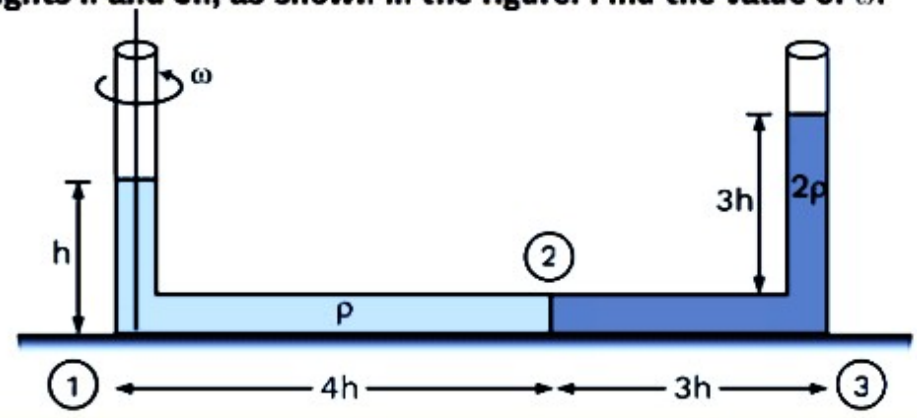
$\Rightarrow \quad dP = (\rho x \omega^2)dx$

$\Rightarrow \quad \Delta P = \int_0^x (\rho\omega^2)\, x dx$

$\Rightarrow \quad \Delta p = \frac{\rho\omega^2 x^2}{2}$

$\Rightarrow$ Increases in pressure at a distance 'x' from the rotational axis is $\frac{\rho\omega^2 x^2}{2}$

Q. **A vertical U-tube is rotated about the vertical axis passing through the left hand limb, with an angular speed ω such that the liquid columns in the limbs have heights h and 3h, as shown in the figure. Find the value of ω.**



Sol. $P_1 = P_0 + \rho gh$ .....(i)

Pressure difference between P_1 and P_2

$$P_2 - P_1 = \frac{\rho\omega^2}{2}\left[(4h)^2 - 0^2\right]$$

$\Rightarrow$ $P_2 - P_1 = 8\rho\omega^2$ ....(ii)

Pressure difference between P_3 and P_2

$$P_3 - P_2 = \frac{2\rho\omega^2}{2}\left[(7h)^2 - (4h)^2\right]$$

$\Rightarrow$ $P_3 - P_2 = 33\rho\omega^2$ ....(iii)

$P_3 = P_0 + (2\rho)g\,(3h)$

$\Rightarrow$ $P_3 = P_0 + 6\rho gh$ ....(iv)

Now by adding equation (i), (ii) ad (iii) and subtracting equation (iv)

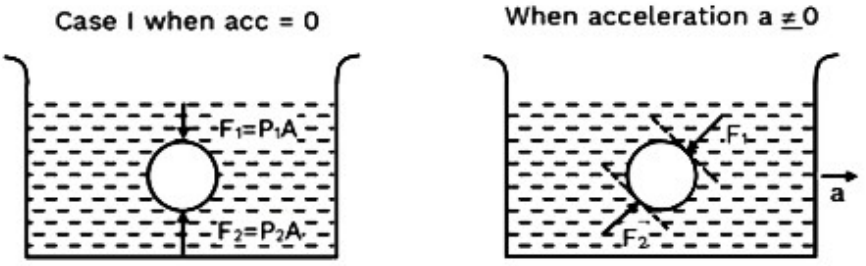
$0 = P_0 + \rho gh + 8\rho\omega^2 h + 33\omega^2 h - (P_0 + 6\rho gh)$

$5\rho gh = 41\,\rho\omega^2 h$

$\Rightarrow$ $\omega = \sqrt{\frac{5g}{41h}}$

BUOYANCY

Buoyant force: When a solid is immersed in liquid completely/partially, liquid exerts some force on solid. Net force exerted by liquid on solid is called buoyant force.



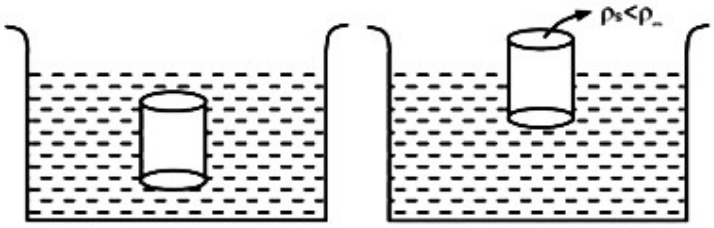
Buoyant Force = $F_2 - F_1$ Buoyant force $F_2 - F_1$

Here ($F_2 > F_1$)

Buoyant Force is because of difference of pressure at different points.

Archimedes' Principle:

Upward buoyant force that is exerted on a body immersed in fluid fully/partially is equal to weight of fluid displaced by body.

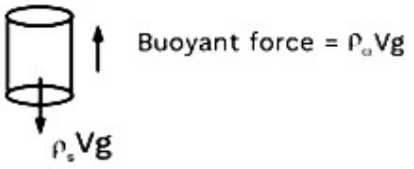


Force by this liquid = weight of water displaced by liquid

$\rho_s > \rho_w \Rightarrow$ complete sink

$\rho_s = \rho_w \Rightarrow$ just sink

$\rho_s < \rho_w \Rightarrow$ partially sink.

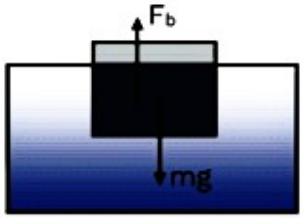


Q. **Find the fraction of an iceberg lying below the surface of the sea water if Density of sea water is (ρ) = 1.028 × 10^3 kg/m^3 and density of ice (d) = 0.917 × 10^3 kg/m^3.**

Sol. For floating conditions, we know

Weight of the iceberg(mg) = Buoyant force (F_b)

Consider V to be the total volume of the iceberg and v be the volume inside the sea water. Then



$$Vdg = v\rho g$$

$$\Rightarrow \quad \frac{v}{V} = \frac{d}{\rho}$$

$$\Rightarrow \quad \frac{v}{V} = \frac{0.917 \times 10^3}{1.028 \times 10^3} = 0.892$$

Q. **A cubical block of wood 10 cm along each side floats at the interface between an oil and water with its lowest surface 2 cm below the interface. If the heights of oil and water columns are 10 cm each and ρ_{oil} = 0.8 g/cc. final (a) the mass of the block, (b) the liquid pressure (gauge pressure) at the lower surface of the block.**

Sol. (a) For the floating cube,

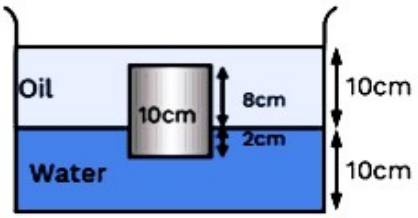
Weight of the cube = Buoyant force

$\Rightarrow$ $mg = F_{B(oil)} + F_{B(water)}$

$= 8 \times 10^2 \times \rho_{oil} g + 2 \times 10^2 \rho_w \times g$

$\Rightarrow$ $m = (800 \times 0.8) + (200 \times 1)$

$= 840$ g



(b) The gauge pressure at the lower surface of the cube is

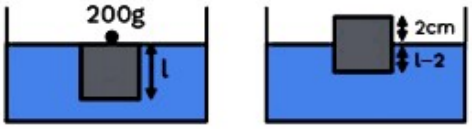
$$P_{oil} + P_w = \frac{10}{100} \times (0.8 \times 10^3) 9.8 + \frac{2}{100} \times (1 \times 10^3) 9.8$$

$= 980$ N/m^2

(Here we have used h_{oil} = 10 cm and h_w = 2 cm above the lower surface)

Q. **A cube of wood supporting a 200 g mass just floats in the water. When the mass is removed, the cube rises by 2 cm. What is the size of the cube?**

Sol. The cube 'just floats' means that it is fully under water while floating.



If m is the mass of cube, then we have,

$(m + 200)g = l^3 \rho g$ ...(i)

When the mass is removed,

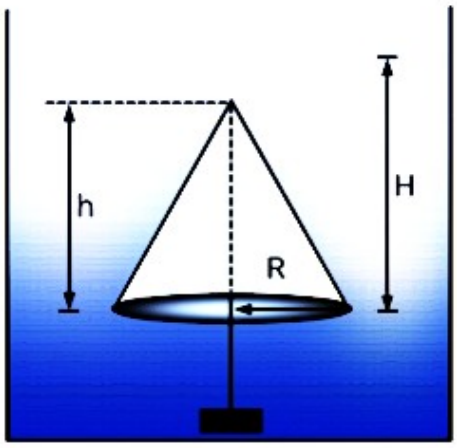
$mg = (l-2)l^2 \rho g$ ....(ii)

Subtracting Eq. (ii) from Eq. (i), we have

$200 = \rho [l^3 - (l-2)l^2]$

$\Rightarrow$ $l = 10$ cm (using density of water ρ = 1 g/cc of water)

Q. **A wooden solid cone of base radius R and height h is tied to a string and kept inside a liquid as shown. If in the equilibrium condition, the depth of the base of the cone below the free surface of water is H, then**



(a) What is tension in the string.
(b) What is the force on the slant surface of cone exerted by water.
(Take atmospheric pressure to be P_0 and Density of wood d (<ρ, density of water).

Sol. (a) Let Buoyancy force is $F_B = V_{cone}\rho g$
Weight of the cone $W = V_{cone}dg$

where $V_{core} = \frac{1}{3}\pi R^2 h$

$\therefore$ Tension $T = W - F_B = \frac{1}{3}\pi R^2 hg(d - \rho)$

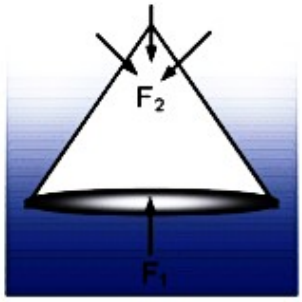
(b) Buoyancy is resultant of vertically upward force (F_1) applied by water pressure on the base of the cone and the vertically downward force (F_2) applied by water on the slant surface. The horizontal component sums up to zero due to symmetry.

$F_B = \frac{1}{3}\pi R^2 h\rho g$

$\therefore$ $F_1 - F_2 = F_B$

$\pi R^2[P_0 + \rho gH] - F_2 = \frac{1}{3}\pi R^2 h\rho g$

$\therefore$ $F_2 = \pi R^2 [P_0+\rho gH] - \frac{1}{3}\pi R^2 h\rho g$



Buoyant force in accelerating fluid

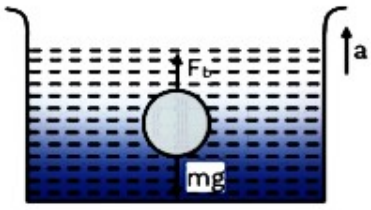
Suppose the body is submerged into a liquid of density ρ, which is accelerating upwards. If a is the acceleration of the liquid, then,

$F_b - mg = ma$

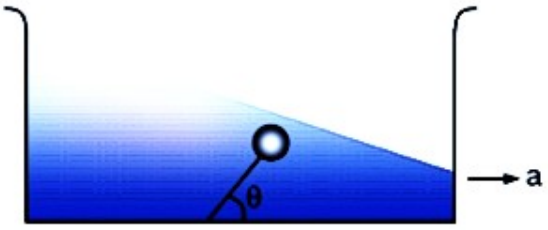
or $F_b = m(g + a)$

Here m is the mass of the displaced liquid, which is equal to Vρ. therefore,

$F_b = \rho V(g + a)$



Q. **A container which is filled partially with water is moving in horizontal direction with acceleration a=g/3. A small wooden ball of mass m is tied to the bottom of the container using a string. The ball remains inside water with the string inclined at an angle θ to the horizontal. Assuming that the density of ball is half the density of water, find the tension in the string.**



Sol. As the container is accelerating, the liquid surface will be inclined with horizontal such that

$$\tan\theta = \frac{a}{g} = \frac{g/3}{g} = \frac{1}{3}$$

Hence $\sin\theta = \frac{1}{\sqrt{10}}$ and $\cos\theta = \frac{3}{\sqrt{10}}$

In this case the buoyant force on the ball will be normal to water surface. The magnitude of buoyant force

$$B = V_{dis}\rho_w g_{eff}$$

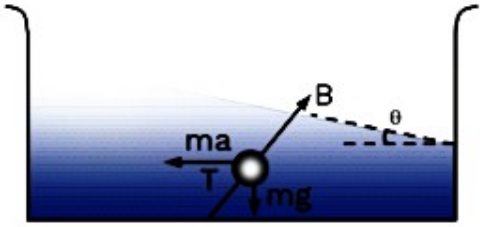
$$\Rightarrow \quad B = \left(\frac{m}{\rho}\right)\rho_w\sqrt{g^2 + a^2} = 2m\sqrt{g^2 + \frac{g^2}{9}}$$

which gives $B = \frac{2\sqrt{10}}{3}mg$

The forces acting on the ball are shown in FBD (w.r.t. container)

As the ball is in equilibrium with respect to container,

then $B = T + mg\cos\theta + ma\sin\theta$

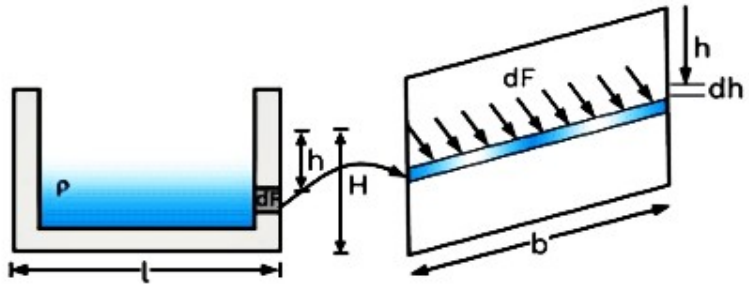


$$\frac{2\sqrt{10}}{3}mg = T + mg\cdot\frac{3}{\sqrt{10}} + m\left(\frac{g}{3}\right)\cdot\frac{1}{\sqrt{10}}$$

$$\Rightarrow \quad T = \frac{\sqrt{10}}{3}mg$$

Force acting on the vertical wall

Pressure acting on the vertical wall is not uniform but increases linearly with depth. Pressure at a depth h from the free surface is $p = \rho gh$.



Force dF acting on a differential element of height dh is

$$dF = p(bdh) = (\rho gh)(bdh) = \rho gbhdh$$

The total force is

$$F = \rho gb\int_0^H h\,dh = \frac{1}{2}\rho gbH^2$$

The total force acting per unit width of the vertical wall is

$$\frac{F}{b} = \frac{1}{2}\rho gH^2$$

The torque on vertical wall is given by

$$\tau = \int_0^H h\,dF.$$

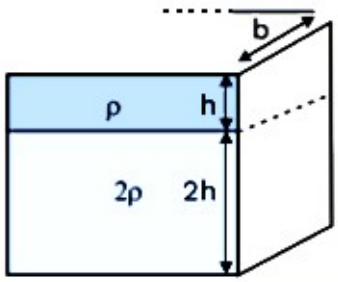
$$\tau = \int_0^H h(\rho g b h\,dh) = \rho g b \int_0^H h^2 dh = \frac{1}{3}\rho g b H^3$$

The point of application of total force from the free surface given by

$$h_c = \frac{\tau}{F} \Rightarrow h_c = \frac{2}{3}H$$

The total force acts at a depth 2/3 H from the free surface.

Q. **Two immiscible liquids of densities ρ and 2ρ and depths h and 2h respectively are pushing a vertical plate as shown. Find the net lateral thrust exerted by the liquids on the vertical plate.**



Sol. If F_1 and F_2 are the hydrostatic forces acting on the rectangular patches of area A_1 and A_2, then the total hydrostatic force on the gate is:

$$F = F_1 + F_2$$

Where $F_1 = P_{av_1} A_1$ and $F_2 = P_{av_2} A_2$.

P_{av_1} and P_{av_2} are the average pressures on the areas A_1 and A_2, respectively.

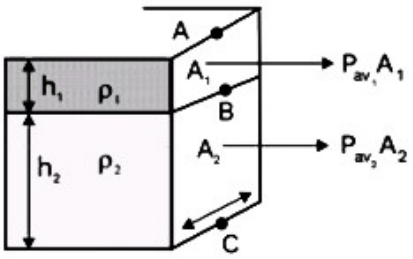
$$F = \rho_{av_1} A_1 + P_{av_2} A_2 \text{ and}$$

Substituting $P_{av_1} = (P_A + P_B)/2, A_1 = bh_1 = bh,$

$P_{av_2} = (P_B + P_C)/2,$

$A_2 = bh_2 = b(2h)$ we have

$$F = \frac{(P_A + P_B)}{2} bh + \frac{(P_B + P_C)}{2} 2bh$$



Substituting $P_A = 0, P_B = \rho gh$

and $P_c = (\rho_1 h_1 + \rho_2 h_2)g = [\rho h + (2\rho)(2h)]g = 5\rho gh$

we have $F = \frac{13}{2}\rho gh^2 b$